

1. Temperature System

Create a Temperature class.

- Private attribute: __temp
 - Temperature cannot be less than -50 or greater than 60
 - Display temperature
-

2. Movie Rating

Create a Movie class.

- Private attribute: __rating
 - Rating should be between 1 and 5
-

3. Student Attendance

Create a Student class.

- Private attribute: __attendance
 - Attendance should be between 0 and 100
-

4. Mobile Volume

Create a Mobile class.

- Private attribute: __volume
 - Volume should be between 0 and 100
-

5. Exam Fee

Create a Student class.

- Private attribute: __fee
 - Fee cannot be negative
-

6. Employee Management

Create an Employee class.

Private Attributes:

__name

__salary

Validation:

- Name should not be empty
- Salary must be positive

Methods:

- update_salary()
 - display_employee()
-

7. Online Course

Create a Course class.

Private Attributes:

__course_name

__duration

Validation:

- Duration must be greater than 0

Methods:

- update_duration()
 - display_course()
-

8. Cricket Player

Create a Player class.

Private Attributes:

__runs

Validation:

- Runs cannot be negative

Methods:

- add_runs()
 - display_runs()
-

9. Library Book

Create a Book class.

Private Attributes:

__book_name

__copies

Validation:

- Copies cannot be negative

Methods:

- issue_book()
 - return_book()
-

10. Flight Ticket

Create a FlightTicket class.

Private Attributes:

__price

__seat_number

Validation:

- Price > 0
- Seat number cannot be empty

11. Bank Account System

Create a BankAccount class.

Private Attributes:

__account_holder
__balance

Methods:

deposit()
withdraw()
check_balance()

Validation:

- Deposit > 0
- Withdraw > 0
- Withdraw <= balance

12. Student Result Management

Create a Student class.

Private Attributes:

__name
__python_marks
__sql_marks
__statistics_marks

Methods:

update_marks()
calculate_total()
calculate_percentage()
display_result()

Validation:

- Marks between 0 and 100
-

13. E-commerce Product

Create a Product class.

Private Attributes:

__product_name
__price
__stock

Methods:

update_stock()
purchase()
display_product()

Validation:

- Price > 0
 - Stock >= 0
-

14. Hospital Patient Management

Create a Patient class.

Private Attributes:

__patient_name
__age
__bill

Methods:

update_bill()

display_patient()

Validation:

- Age between 0 and 120
 - Bill cannot be negative
-

15. Employee Payroll System

Create an Employee class.

Private Attributes:

__employee_name

__basic_salary

__bonus

Methods:

add_bonus()

calculate_salary()

display_salary()

Validation:

- Salary > 0
 - Bonus >= 0
-

Advanced Real-World Challenges

16. ATM Machine

Features:

- Private PIN
- Private Balance

Methods:

deposit()
withdraw()
change_pin()
check_balance()

Validation:

- Correct PIN required
 - Withdraw $\leq$ Balance
-

17. Netflix Account

Private Attributes:

__email
__password
__subscription_plan

Validation:

- Email contains "@"
- Password length ≥ 8

Methods:

change_password()
upgrade_plan()
display_account()

18. Laptop Store

Private Attributes:

__brand
__ram
__price
__stock

Validation:

- RAM in [4, 8, 16, 32, 64]

- Price > 0

Methods:

sell_laptop()

restock()

display_details()

19. School Management System

Private Attributes:

__student_name

__roll_number

__fees

Methods:

pay_fees()

remaining_fees()

display_student()

Validation:

- Fees cannot be negative
-

20. Mini Banking Project

Create a complete banking system.

Private Attributes:

__account_number

__account_holder

__balance

Methods:

deposit()

withdraw()

transfer()

check_balance()

Validation:

- Deposit > 0
- Withdraw > 0
- Transfer amount $\leq$ balance